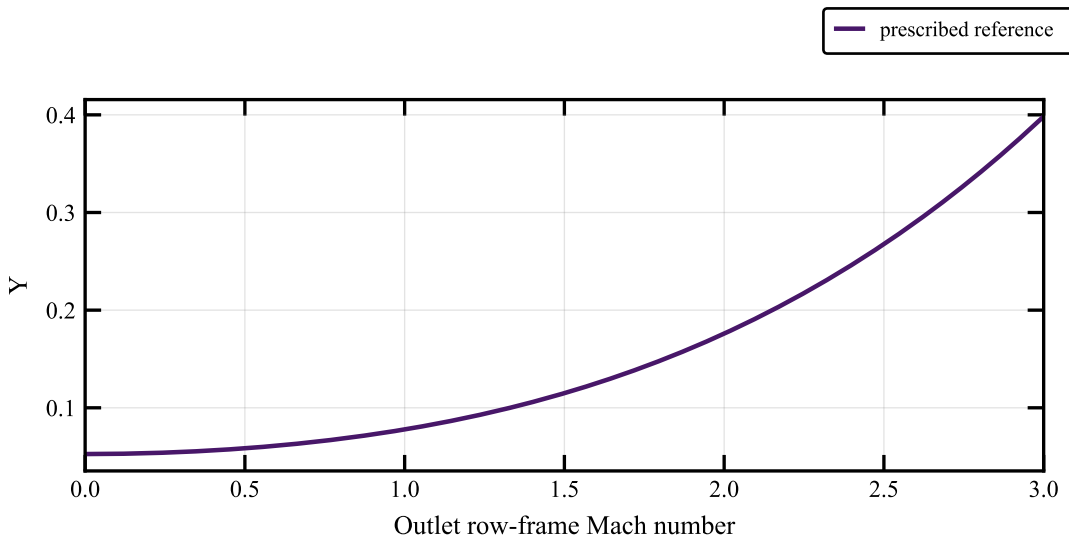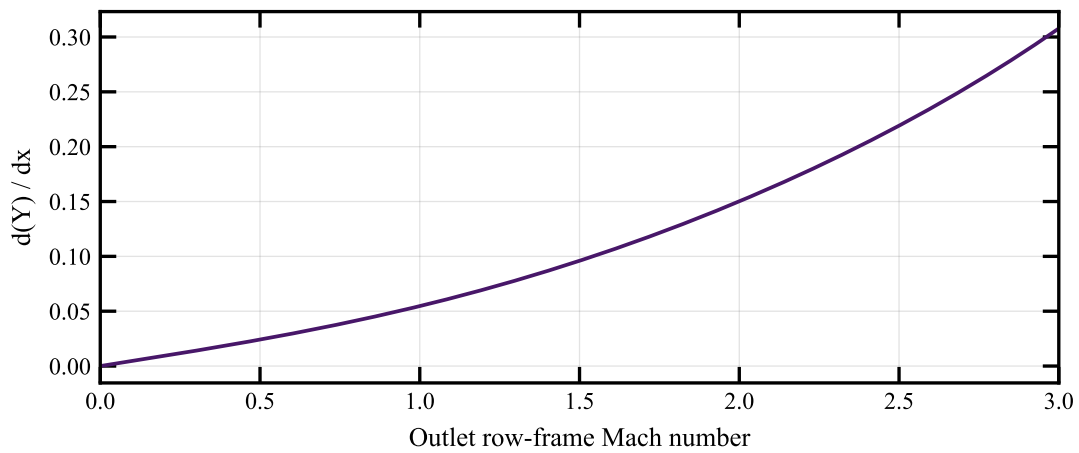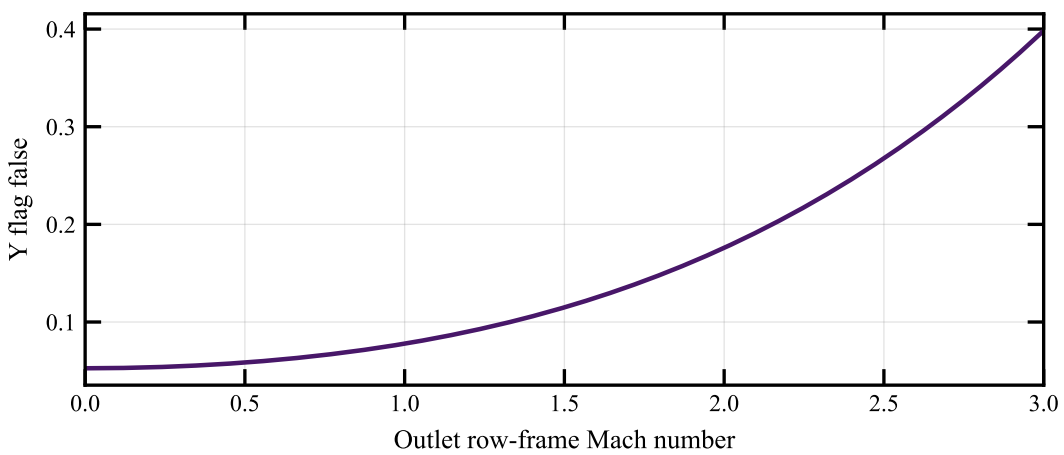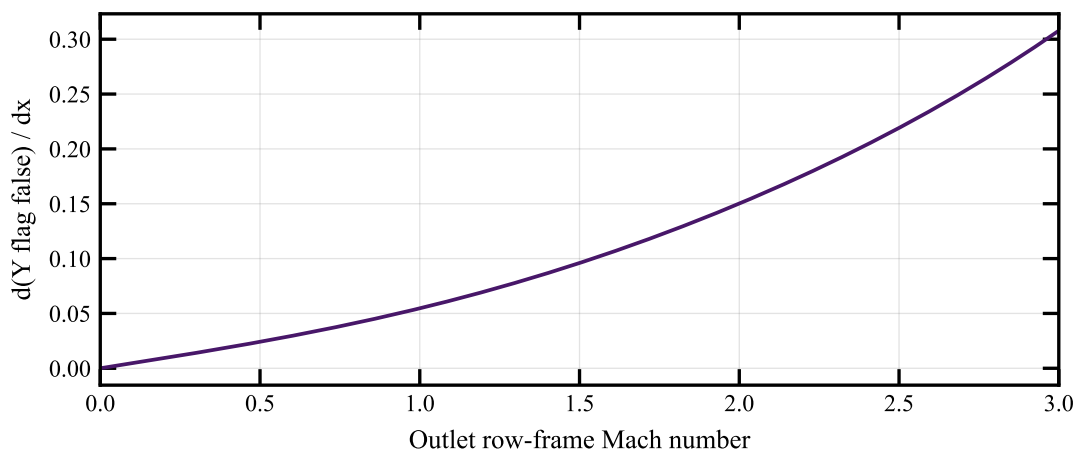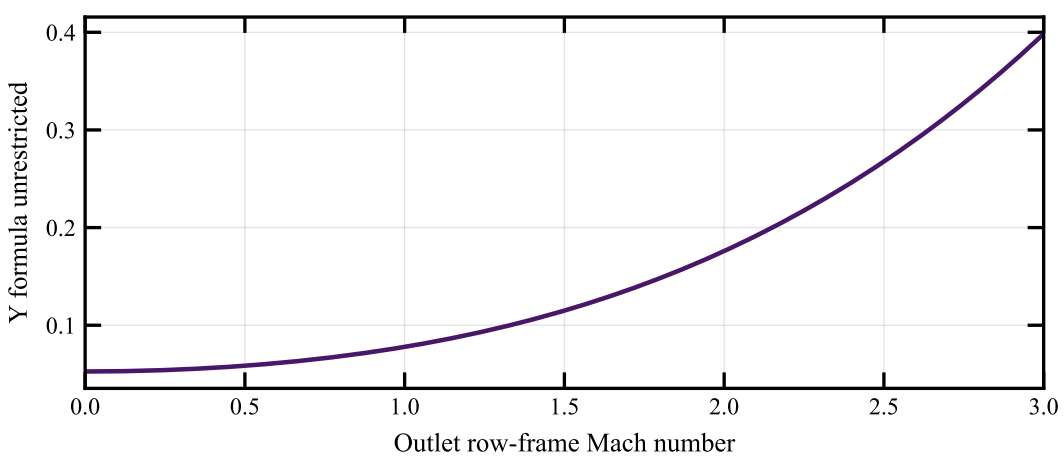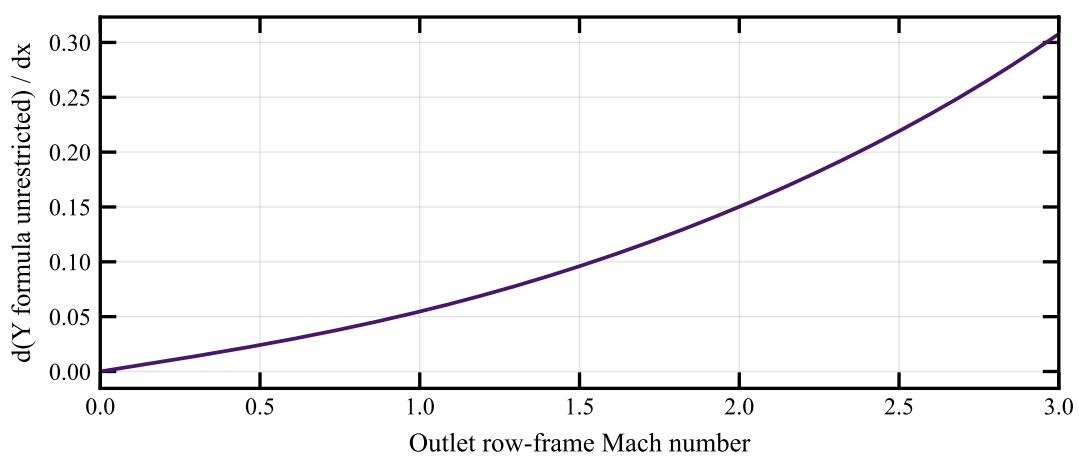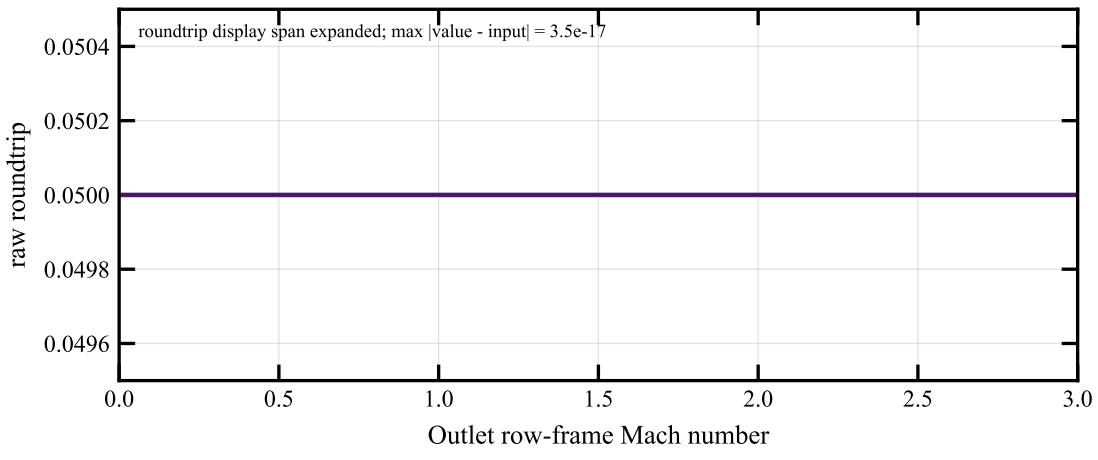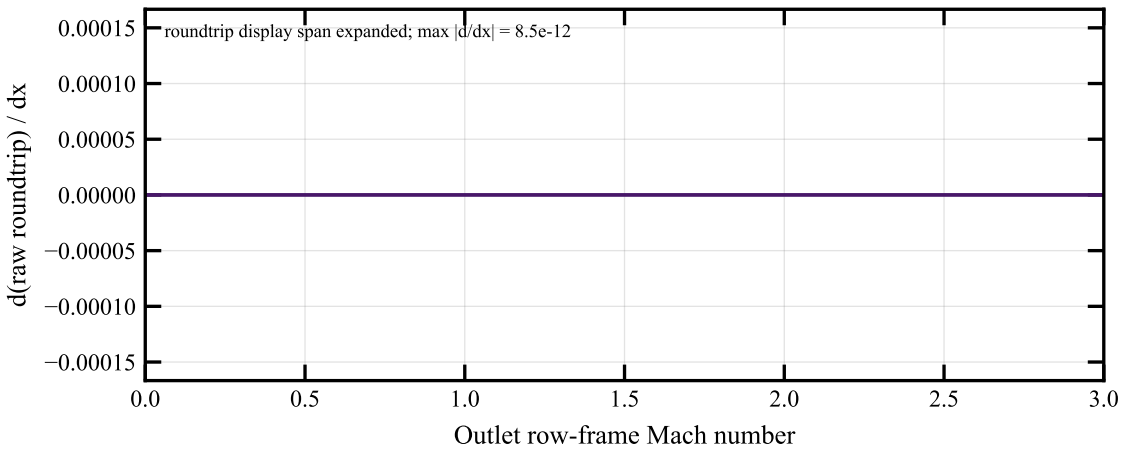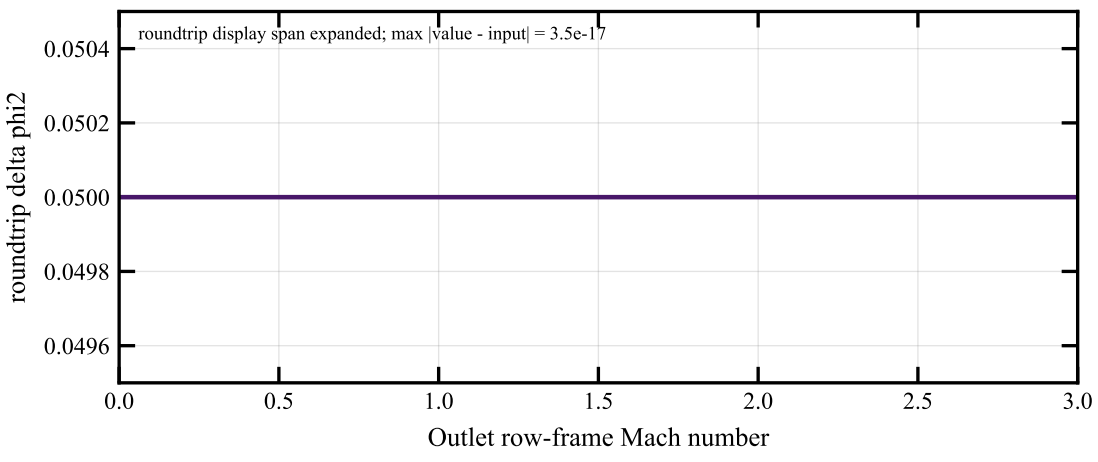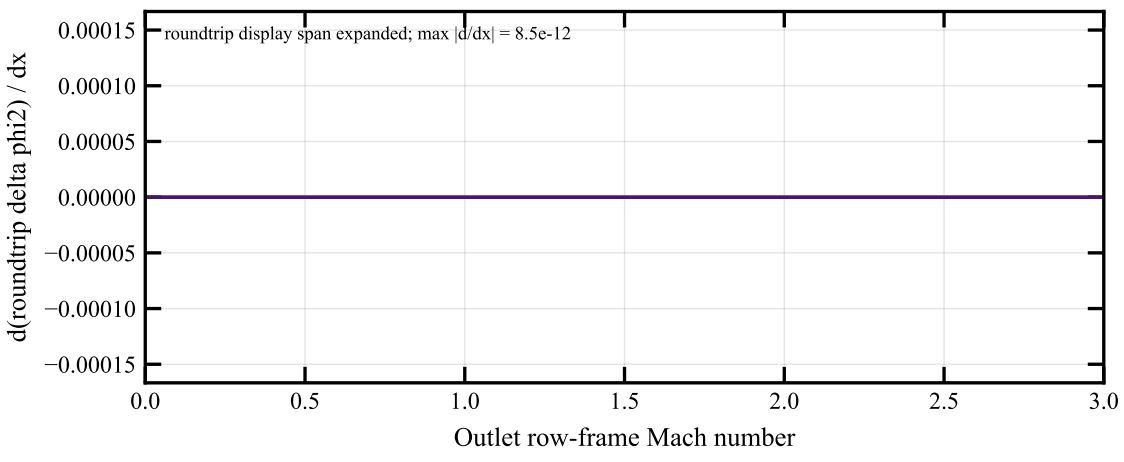conversion mach d0.05 — values and first derivatives